

Enabling Environment and Digital Technology Use Readiness in Agriculture in the Zambezi Riparian Countries

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ABSTRACT

This study sought to understand the extent of digital technology usage across Zambezi riparian countries and analyze whether there is a conducive enabling environment for digital technology upscaling for climate adaptation. The study thus, examined the relationship between having a better enabling environment and the use of digital technologies in agriculture. Using data collected through stakeholder consultations in Malawi, Mozambique, Zambia and Zimbabwe as well desk studies focusing on the rest of the Zambezi riparian countries (Angola, Botswana, Malawi, Mozambique, Namibia, Tanzania, Zambia and Zimbabwe, the study finds that having an enabling environment for the upscaling of digital technologies varies from one Zambezi riparian country to the other, with Botswana and Namibia, having a better enabling environment and, thus a better readiness for digital technology upscaling. Malawi, Mozambique, and Tanzania have enabling environments that beg for improvement to facilitate large scale adoption and upscaling of digital technology. Other key constraints that undermine the adoption and use of digital technologies across the Zambezi River Basin countries are related to availability, access, affordability, technology design, ease of doing business, cultural and family norms, gender, and poor user skills. The cost of data was observed to be one of the major constraints for digital technology use in climate change adaptation across several countries. Macroeconomic conditions, particularly inflation and lack of foreign exchange, also undermine growth prospects of digital technology adaptation as many of these technologies need to be imported. It is interesting that correlations between enabling environment indicators and digitalization variables appear to reveal a positive relationship signifying that to upscale the use of digital technologies in agriculture, it is critical to address parameters within the enabling environments including, pre-requisite infrastructure, macro-economic stability, policy, and regulatory environment.



1. Introduction

The Zambezi River Basin (Figure 1) is the largest in the Southern African Development Community, and the fourth largest basin in Africa, with a total area of 1.37 million square kilometers (World Bank 2010). It is shared by eight countries including Angola, Botswana, Malawi, Mozambique, Namibia, Tanzania, Zambia, and Zimbabwe. It has high transboundary

importance in terms of water management. This is not only because of the different contributions of sub-basins to the overall runoff of the Zambezi River but also because of the very high variability of climate change, the valuable natural resources, and major differences in the sub-basin characteristics. Its waters are critical to sustainable economic growth and poverty reduction in the region. In addition to meeting the basic needs of some 30 to 40 million people and

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sustaining a rich and diverse natural environment, the river plays a central role in the economies of the eight riparian countries (Zambezi River Authority 2007). Eighty percent of the people in the basin depend on the river for agriculture and fishing (Sanchez 2018). The Zambezi River also supplies 20 percent (and likely to rise to 40 percent) of its total water runoff for major economic activities including hydropower generation, irrigation, and both urban and rural water consumption (Hamududu and Killingtveit 2016).

1.1. Poor climate change outlook in the Zambezi region, with low levels of digitalization

The Zambezi River Basin is characterized by extreme climatic variability and climate change. Climate variability has subjected the river and its tributaries to a cycle of floods and droughts that have devastating effects on the people and economies of the region, especially the poorest members of the population (Brazier 2015). One of the key impacts of climate change in the Zambezi region is altered rainfall patterns. Changes in precipitation can result in both droughts and floods, affecting agricultural productivity and water availability. According to the IPCC's Fifth Assessment Report (2014), southern Africa, including the Zambezi region, is projected to experience a warming trend of 1.5 to 3 degrees Celsius by the end of the century. Increased temperatures can exacerbate water scarcity and lead to higher evaporation rates, further impacting water resources in the region. Additionally, the Zambezi region is home to diverse ecosystems and climate change threatens these ecosystems, impacting biodiversity, river flow patterns, and the overall ecological balance. Changes in river flow can have cascading effects on wildlife, aquatic species, and the livelihoods of local communities dependent on these resources. This is exacerbated by the rising population in the basin which is projected to result in an increase in water demand over time, which in-turn, will likely put more pressure on water allocation among the eight riparian countries (Beyer et al. 2016; Beck and Bernauer 2011). It is thus imperative that efforts are made towards adopting interventions that will support the Zambezi River Basin in adapting to projected changes that will impact livelihoods.

The adoption of digital technology has been identified as critical in facilitating climate adaptation in the agriculture sector (AfDB 2021; Kling et al. 2014). Digital technology plays a key role in adaptation including influencing the behaviors of farmers, input providers, processing and retail companies, and marketing companies. The adoption of digital



Figure 1 The Zambezi River Basin and its subsidiaries (Source: Senzanje and Dirwai, 2020)

tools in agriculture improves efficiency along the production chain, improves access to information that can guide decision making, and enables virtual transactions (Mayoyo et al. 2023). However, digital technologies in the basin are not widely used and their potential, in the context of agriculture, is not yet well understood. Adoption of digital technology still faces limits and barriers to adaptation in the agriculture sector.

These limits can be natural, technological, economic, social, or institutional. Natural limits range from ecosystem thresholds to geographical and geological limitations. The economic, social, and institutional barriers form key enabling environment barriers and include unintegrated and an outdated policy environment, limited agriculture financing, lack of infrastructure, and technology gender divide in access to technology among others (Malambo et al. 2023; POTRAZ 2021). A country's enabling environment plays a crucial role in the adoption and use of digital technologies in climate change adaptation in the agriculture sector within the Zambezi River Basin. A lack of a good enabling environment constrains digital technology and adoption and thus use (Fuglie et al. 2019). Stakeholders in the countries, including farmers attest to the fact the enabling environment has been one of the key factors stalling digital readiness. Table 1 below summarizes some of the barriers to adaptation of digital technologies that revolve around enabling environment as found in the course of this research.

Table 1 Barriers to adaptation of digital technologies (Source: Adapted from Malambo et al., 2023, and Mayoyo et al., 2023)

Technological	Infrastructure	Skills /Capacity
• Cost of technology • Affordability of smart devices and other ICT products Limited local ICT innovation, • Research and Development and entrepreneurship • Unavailability of digital technologies • Limited access to technological infrastructure gadgets	• Lack of business entities developing the digital economy • Interoperability challenges Inadequate communications infrastructure (e.g., water, internet, electricity, road and Rail networks) e.g., presence of grey areas without network/radio coverage • Network connectivity challenges because the coverage is still growing • Restricted access to technologies such as satellites	• Inadequate investment capital and financing to scale adoption of climate change adaptation digital tools • Lack of appropriate policies to support ICT sector growth and climate adaptation • Level of Education, ICT Knowledge and Skill of individuals. Low digital literacy among this critical mass. • Women have less access to the internet than men • Credit and insurance constraints lack of confidence in developing and adopting local technologies • Weak agriculture insurance

1.2. Paucity of studies in this area

The paper sought to understand how an enabling environment can either upscale or constrains digital technology adoption and use. The paper thus examines how the enabling environment in the eight riparian countries affects the use of digital technologies in climate change adaptation in agriculture. Literature on this specific area is limited shows that there needs to be more research in this area especially with respect to the role of the enabling environment for digital technology upscaling. Constrained investments by governments, the private sector, and development partners have a bearing on digital technology use in the agriculture sector.

There are limited studies to support and fully guide policymakers when making policy decisions on digital climate change adaptation strategies and policy options. This results in non-robust interventions, of a short-term nature and exhibiting time inconsistency, being employed rather than well-crafted long-term solutions which may be more positively impactful for society in the long term. This paper thus aims to add to the narrow pool of studies that focus on the role of the enabling environment on digital penetration/use in the Zambezi.

2. Literature Review

2.1. Background

Climate change has continued to affect various sectors in the eight riparian countries under study including agriculture. These effects are expected to continue beyond the next thirty years. Projections show that annual temperature in Southern Africa in the next 50 years are likely to increase by 1.5 to 2°C

by midcentury (USAID 2015). This is confirmed by a study in Zambia by Hamududu and Ngoma (2019) that found that all regions within the country would experience temperature increases exceeding 1.5°C by 2050, especially during the growing seasons. This suggests that the Southern and Western regions of Zambia are the most vulnerable to climate change and will likely experience more reduced rainfall over time and in the worst-case scenario, rainfall may reduce by 20–30 percent in these regions. On the contrary, the Northern region was projected to receive nearly 5 percent more rainfall by 2050. The results are also applicable to three other countries including Malawi, Mozambique, and Zimbabwe. USAID (2015) also looked at the rainfall trends up to 2059 and noted that at the regional level, the mean annual temperature will increase are 1.5°C, 1.7°C, and 2.1°C by 2065. Droughts are also expected to increase over the same period.

In instances where rains occur, floods are also expected to occur more in the southern, western, and eastern regions of Southern Africa as well. These predicted changes are expected to affect various production systems including yields (Devenish et al. 2023; Hamududu and Ngoma 2019; USAID 2015). On average, climate change is likely to reduce the yields for maize, other cereals, root crops, and tobacco with appreciable differences in the impacts of climate change on these crops by 2050 (FAO 2015). In addition, maize is projected to be the worst affected crop, with estimated yield reductions of up to 3–6 percentage points in the worst affected Southern and Western regions of Zambia. In the aforesaid regions, yield variability is high, ranging from 26 to 30 percent. Further, cotton and root crops are projected to be less adversely affected by climate change. Cotton yields will increase by 1 percent in the Northern regions of Zambia by

2050, while root crop yields rise by 0.50 percent. These effects on yield are further going to impact other downstream players within the various value chains such as small businesses and Small and Medium Enterprises (SMEs) who may not be able to purchase the optimum amounts of outputs for processing and/or onward trading (Ngoma et al. 2021).

The projected climate change features of Southern Africa are summarized as follows (Hamududu and Ngoma 2021; Mtetwa 2018; USAID 2015; World Bank 2019; UNDP 2022; World Bank 2019b; Midgley 2011):

- Increase in temperature of 1.2–2.2°C between 2040 and 2059.
- Increase in days with maximum temperatures of 35°C by 39 days between 2040 and 2059.
- More variable precipitation, with some models projecting an increase in the long term, and others projecting a decrease.
- Increased rainfall in the north and east; reduced rainfall in the south and west.
- Dry spells lasting 13 days longer than the average current duration between 2040 and 2059.
- 21 percent increased incidence of severe drought between 2040 and 2059.

Digital technologies (such as mobile devices and social media, Precision agriculture and remote sensing technologies, blockchain, financing and insurance systems and intelligence systems, among others) can be crucial adaptation strategies in the face of climate change shocks, particularly reduced rainfall, and water access tension. The technologies can be applied to all aspects of agrifood systems and reflect a change in generalized management of resources towards highly optimized, individualized, intelligent, and anticipatory management, in real time, hyper connected and driven by data (Trendov et al. 2019). There are still opportunities to increase and cascade the benefits of digital technologies in climate adaptation in agriculture to efficiently use water resources in periods of scarcity.

2.2. Conceptual framework for digital technology adoption and upscaling

This conceptual framework takes on a Theory of Change approach that is expected during the transformation process as farmers adopt digital technologies (Figure 2).

Figure 2 shows that digital technology adoption is influenced

by several factors, directly and indirectly. An enabling environment, for example, is of paramount importance in influencing the adoption and use of digital climate adaptation technology. Key elements of the enabling environment include the current levels of essential infrastructure required to sustain digitalization investments: Electricity, middleware infrastructure such as mobile towers, other technologies, mobile phones, as well as computers (FAO and ITU 2022). The other vital element that affects the enabling environment is the current level of digital penetration, which refers to the availability of digital services in the agricultural sector. Here, factors impacting the digital agriculture transformation process within a country, including internet users, active subscriptions, cost of mobile phones, digital finance, among others, are considered important to determine whether digital technologies will be adopted widely or not. This includes digital analytics, middleware infrastructure, hardware as well as software.

Policy and regulatory environment of the Zambezi River Basin countries as well as of their respective regional economic communities such as the SADC, and indeed the African Union, is also a key element for an enabling environment. This encompasses the levels of e-Government services in a country, as well as existing mechanisms and institutional support, policies, laws and regulatory frameworks that are needed to nurture and support digital agriculture transformation (FAO and ITU 2022). They are also vital in influencing availability, costs, and hence the adoption and use of technologies. Finally, the general business environment is also as important as the regulatory environment. For example, whether the country scores better on ease of doing business is a decisive factor as it may approximate the attractiveness of the country for digital technology businesses. Aside from the enabling environment, factors at the technology, adopter, and the environment (population) level are also significant.

2.3. Role of digital technology in adaptation

Digital technology plays a key role in adaptation including influencing the behaviors of farmers, input providers, processing and retail companies, and marketing companies. Digital technologies have been very instrumental in supporting farmers' decision making based on information received to improve preparedness in the face of climate hazards. Data generating tools such as early warning systems have been used to map the impending disaster for example drought and the messages are being conveyed through digital tools such as mobile phones, radio, TVs, newspapers, social media platforms and as a result, this has improved disaster

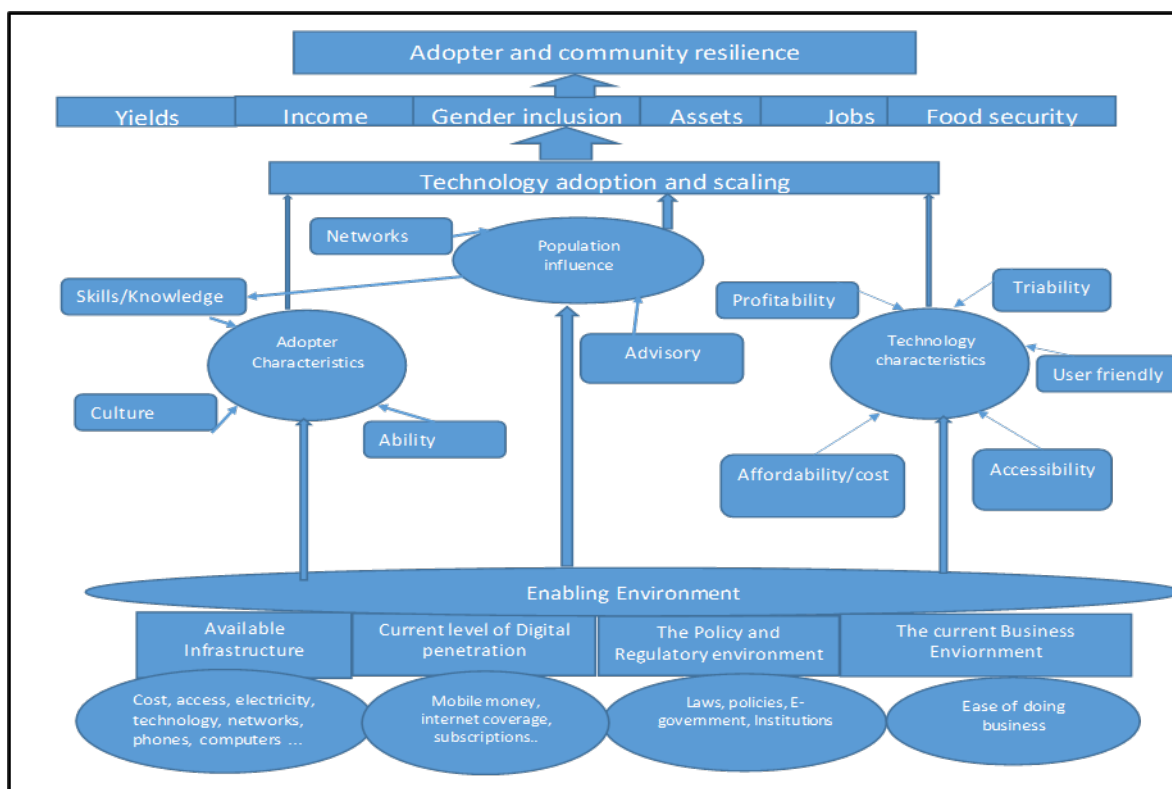


Figure 2 Digital technology adoption, scaling, and resilience (Source: Authors' illustration based on Matchaya 2024, *forthcoming*)

preparedness (Malambo et al. 2022, FAO 2016).

Adoption of digital tools in agriculture improves efficiency along production chain, improves access to information that can guide decision making and enable virtual transactions (Mayoyo et al. 2023). Specific examples of digital technology applications in agriculture include digital and wireless technologies for data measurement, weather monitoring, animal monitoring, geospatial monitoring and precision application of water and chemicals. Satellite imagery in cropping is also another digital technology. Digital infra-red light and heat sensors, when used with geographical system technology in drones can be used to measure paddock crop health, and inform the decision on irrigation, pest management, fertilizer application and harvesting (Dinesh and Rodriguez 2019; Goh 2022; CGIAR 2019; Stephenson et al. 2021).

Remote sensing and GIS mapping may help small-scale farmers to use water resources more efficiently and thus lessen water stress. Digital technology can revolutionize farming practices and agri-food value chains in the future. It can

positively disrupt impact on the adaptive capacity of farmers and agribusinesses. Also, digital technologies can help improve production efficiency and effectiveness for smallholder farmers and provide governments with a wealth of data for policy formulation (or adaption) and planning for climate adaptation (Mayoyo et al. 2023). McKinsey (2020) estimates that increasing connectivity in agriculture through digital technologies could add US\$ 500 billion to global GDP by 2030.

Although digital technology has been identified as key to responding to the key climate change impact threats that the Zambezi River Basin faces, a few barriers still exist that stall adaptation progress. The basin, continues to face challenges with unintegrated and outdated policy environment as well as the lack of agriculture financing, undeveloped infrastructure and technology gender divide in access to technology (Malambo et al. 2023; POTRAZ 2021).

3. Methodology

To understand the role of the enabling environment in

influencing the use of digital technology in climate adaptation in agriculture in the Zambezi River Basin countries, qualitative methods were used to collect and analyse data for this study. The researchers conducted desk reviews of the literature on climate risks, adaptation strategies, available digital tools for climate adaptation, country preparedness for digital adaptation, and the opportunities and constraints for scaling up those technologies in each of the eight riparian countries.

The conceptual framework presented in section 2.2 was used as a guideline for the information the researchers sought for analysis throughout the study. The literature informed gaps and opportunities for both public and private sector to invest in digital technologies. It also helped in identifying constraints in the use of digital technologies in climate change adaptation. This was done to explore the correlation between an enabling environment and digital technology adoption and use.

Stakeholder consultations were another key component of the study approach. At least eight multi-stakeholder consultation workshops involving a total of approximately two hundred (200) participants were held throughout the assessment to gather information on climate risks, climate adaptation strategies currently in practice by stakeholders, digital technologies in use, enabling environment, opportunities, constraints, and recommendations for intervention. Specifically, the research team conducted national consultations workshops in Zambia, Zimbabwe, Malawi, and Mozambique. Regional meetings in Maputo, Mozambique and Lilongwe, Malawi served as a platform for validating the findings of the assessment reports at the Zambezi regional level.

The stakeholder consultation workshops drew diverse stakeholders, carefully selected to represent the various interest groups existing within the Zambezi River Basin countries. All consultation workshops comprised farmer representatives including women and youth farmers, government, civil society, and private sector groups. The outcomes of the stakeholder consultation workshops, which comprised of information on climate risks, lists and inventories of digital technologies used in climate adaptation, opportunities, constraints, and recommendations, were then integrated into the national profiles and thereafter the regional assessment report. Backed by the literature review, secondary data analysis results alongside the outcomes of the consultation workshops provided better insights into the key issues around digital adaptation in agriculture within the Zambezi River Basin countries. Key informants provided a

more technical perspective on available digital climate change adaptation.

3.1. The Digital Adaptation Readiness Index and implications

At regional level, an index was developed to track the readiness for digital upscaling in the Zambezi River Basin. The index, called the Digital Adaptation Readiness Index (DARI), has seven pillars identified from the theory of adoption as presented in Section 2.2: adopter characteristics, technology characteristics, business environment, policy and regulatory environment, e-extension, essential infrastructure, prior digital penetration, and relevance of agriculture. Each of these pillars has indicators as shown in Table 2.

3.2. Methodology for computing the baseline Digital Adaptation Readiness Index

Normalization and weighting of indicators

Raw data and sub-indicators were used to calculate composite indices. Indicators were normalized for comparability (Nardo et al. 2005; Saisana and Tarantola 2002). Using the min-max normalization method, the individual indicators were re-scaled from 0 to 100, with 0 being the lowest (worst) performance and 100 the highest (best) possible score. The min-max normalization method is also widely used in other composite indicators such as the Gender Development Index, the Human Development Index (UNDP 2016), and the SDG Index (Sachs et al. 2016).

Explicit weighting (equal weighting, expert weighting), and statistical weighting (regression analysis, factor analysis) are some of the methods used in the development of composite indices and indicators (Nardo et al. 2005). Weighting impacts on the outcome of the composite indicator as well as the ranking of country performance therefore requires important considerations. This article applies equal weighting similar, in line with previous research in other fields (Sachs et al. 2016; Schmidt-Traub et al. 2017).

The various indicators were then normalized to values between 0 and 100 for purposes of developing the Digital Adaptation Readiness Indicator. The normalization was implemented using the Min-Max principles in which the transformed value, for example, takes the form of Equation 1:

$$x'_{ic} = 100 * \frac{x - \min(X)}{\max(X) - \min(X)} \quad (1)$$

Table 2 Pillars and indicators of Digital Adaptation Readiness (Source: Authors)

Pillars	Indicators	Measure	Effect of increase on readiness
Adopter level	Adult male literacy	%	+
	Youth literacy	%	+
	Gender gap in literacy	%	-
	Agricultural land per total	%	+
	Income per capita	US\$	+
	Alternative income	3; more; 2 less; 1 negligible	-/+
Technology Characteristics	Cost of complimentary technology	Affordability (internet, mobile bundles) %	+
Business environment	Cost of borrowing	Lending interest rate %	-
	Ease of doing business	index	+
	Openness to trade	Trade % of GDP	+
	Macroeconomic stability	Inflation %	+
Policy and regulatory environment	Public administration rating	1 Lowest 6 Highest	+
	Property rights and rule-based governance	1 Lowest 6 Highest	+
	Fiscal policy rating	1 Lowest 6 Highest	+
	Availability of digitization policy/strategy and laws	0, Absent, 1 In Progress, 2 Good Progress, 3 Yes	+
	Availability of climate adaptation plans/strategy	0 No; 1 Yes	+
E-extension	e-government	1 is poor; 100 is good	+
	e-participation	2 is poor; 100 is good	+
	Network readiness	3 is poor; 100 is good	+
Essential supportive infrastructure	Rural people with electricity	% of rural	+
	Mobile network coverage	% of population	+
Prior digital penetration	Mobile subscribers per 100	Individuals with subscriptions %	+
	Internet penetration per 100	Individuals using internet %	+
	Computers at home	Households with	+
Relevance of agriculture	Relevance of agriculture (e.g., share of agriculture to total GDP)	%	+/-
	Share of pop in primary agriculture	%	+
	Share of women population in primary agriculture	%	+

This applies to values for indicators where more is good for digital upscaling. For indicators such as costs of technology, more is bad. These indicators were transformed so that high values scored less, and the formula used is represented in Equation 2:

$$x''_{ic} = 100 * \frac{Max(X) - x}{Max(X) - \min(X)} \quad (2)$$

Calculation of the Digital Adaptation Readiness Index

The composite indicator DARI for each country c is usually a simple linear weighted function of a total of N normalized sub-indicators x''_{ic} with weights w_i can be calculated using Equation 3:

$$DARI_c = \sum_{i=1}^N \frac{w_i x''_{ic}}{w_i} \quad (3)$$

Where x''_{ic} is as defined in (1) or (2) as the case may be, and represents the value of the indicator i for the country c at the time t , w_i represents the weight for indicator i .

The calculation of the agriculture-related DARI for the Zambezi was based on the arithmetic mean method of aggregation. This method has been widely applied in the calculation of composite indices such as the United Nations Human Development Index (UNDP 2016), Gender Development Index (UNDP 2016), baseline African Green Growth Index (Kararach et al. 2017), Global Innovation Index (Cornell University et al. 2016), SDG indicator and dashboards for countries (Sachs et al. 2016; Schmidt-Traub et

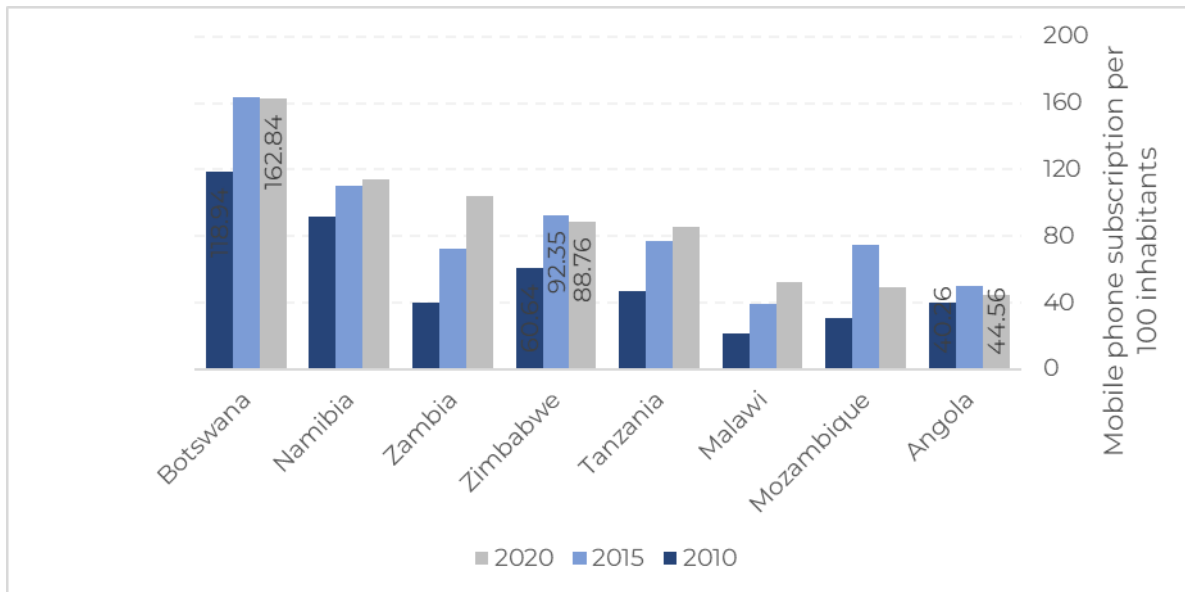


Figure 3 Mobile phone subscription per 100 inhabitants in 8 riparian countries of the Zambezi in 2010-2020 (Source: ITU)

al. 2017; Nhemachena et al. 2018). Other functional forms generally used for aggregation in calculating composite indices are the geometric average and the Leontief production function (Sachs et al. 2016). The arithmetic mean method is widely used compared to the other methods for its ease of application and communication. Therefore, the composite index was generated by averaging with the arithmetic mean, assuming equal weighting across indicators.

4. Findings

Most countries in the Zambezi basin have a wide digital coverage albeit of basic digital infrastructure. The regional average ease of doing business score was 51.8 on a scale of 0 to 100, below the OECD high-income average of 78.4 and the global average of 63.0 (World Bank 2020). This implies that there are still many barriers to doing business in climate adaptation including policy, regulatory, infrastructural and access to finance among others. However, some countries are doing better than others.

A comparative analysis of mobile phone subscriptions for example, between 2010 and 2020 in the riparian countries of the Zambezi shows that Botswana had the highest mobile penetration rate of 162.84 mobile phones per 100 inhabitants in 2020 compared to 118.94 in 2010. Angola had the least penetration rate of 44.56 mobile phones per 100 in 2020 from 49.79 mobile phones in 2015. In Zimbabwe mobile

penetration rate stood at 88.76 mobile phones per 100 people in 2020 compared to 92.34 in 2015, a reversal in the gains made in 2015 (Figure 3) (World Bank 2022; World Bank 2020; World Bank 2018; FAO and ITU 2022).

With regards to internet penetration, most countries fared favorably against other regional countries, but the penetration rates remained low averaging at 70 percent (ZimStat and POTRAZ 2020). This low penetration presents challenges associated with adopting digital technologies that use the internet to link up with households and farmers (POTRAZ 2021). shows the calculated overall composite index of digital adaptation readiness in the Zambezi River Basin countries.

In the constructed index of adaptation readiness (Figure 4), Angola, Malawi, and Zimbabwe were least digital adoption ready. The implication is that the factors that contribute to this outcome should be understood for each country to design appropriate interventions while also ensuring an environment that spurs digital technology adoption. The figures below present the relative scores for each of the eight areas in which a country must improve to perform better on the readiness index. In relative terms, Botswana ranked highest in readiness for digital adaptation (Figure 4). Countries such as Botswana scored highest because they have several well aligned factors that count in its favor for digital readiness adaptation.

Figure 5 shows that among the factors that contribute towards

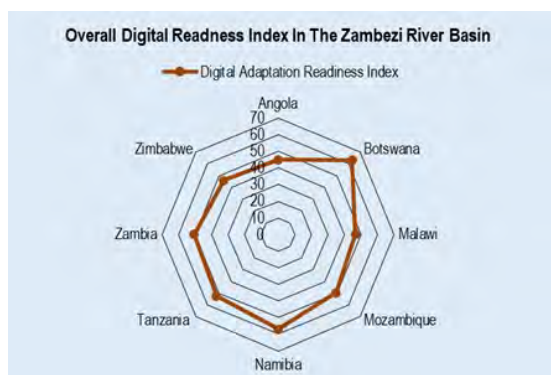


Figure 4 Overall digital adaptation readiness in the Zambezi River Basin countries (Source: Authors)

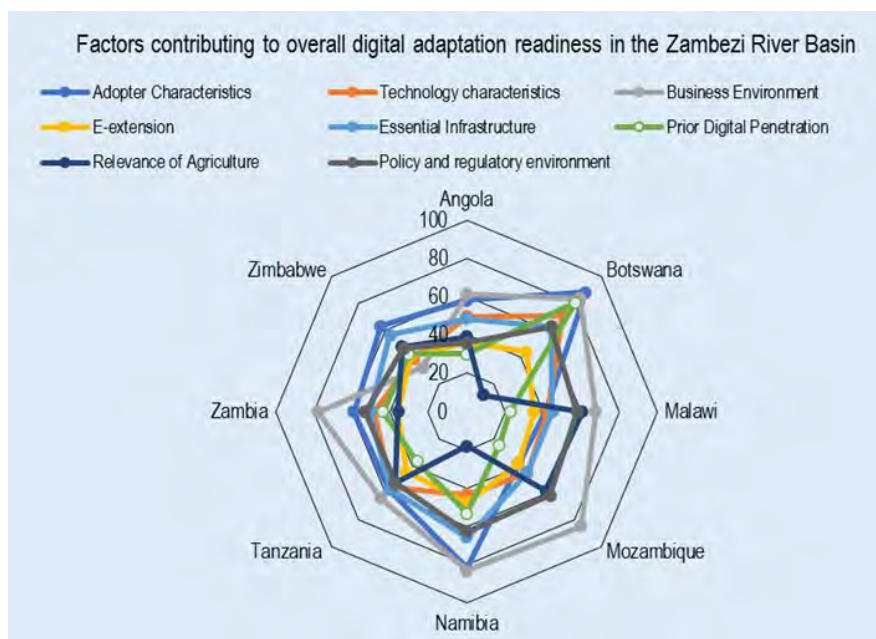


Figure 5: Major factors contributing to overall digital adaptation readiness in the Zambezi River Basin countries (Source: Authors)

digital readiness, an enabling environment scores highly. Botswana scored highly on enabling environment, followed by Mozambique, Namibia and then Zambia. These countries also score highly on the digital readiness as shown in Figure 4. We thus posit that a good enabling environment has a positive correlation to digital readiness. Where a positive enabling environment exists essential infrastructure, policy environment and digital penetration are also be positive. Several factors contribute to the enabling environment including cost of borrowing, ease of doing business, openness to trade and macroeconomic stability. For example, the international Monetary Fund (IMF) projected that the country's conditions remained conducive to credit growth owing to the stability in other parameters within the enabling environment. This has a positive bearing towards the digital readiness index (IMF 2023).

Other conditions that add to an enabling environment include the policy and regulatory environment. Policy and regulatory environment include (Table 2) public administration rating, property rights and rule-based governance, fiscal policy rating, availability of digitization policy/strategy and laws and availability of climate adaptation plans/strategy. Malawi, Zambia Zimbabwe for example, have intensified promotion of climate-smart agriculture practices, establishment of early warning systems for weather-related disasters and the

mainstreaming of climate change considerations into national policies. Such as a National Climate Change Response Strategy (World Bank 2020; World Bank 2022).

Zimbabwe however, scored the least on the index. The factors that pulled down it's score on digital readiness include the ease of doing business with 55 points pointing to an unfavorable business environment and inflation at 314.5 Percent (Statista 2020). The figures below discuss these factors for each country.

Angola is among the top three least ready countries on the overall index of digital readiness for adaptation to climate change. In comparing the performance across the eight pillars to measure digital readiness in Angola, it is clear that the business environment a contribute more to the overall index; however, prior digital penetration (mobile internet signal coverage, subscriptions, computer possession at home), e-extension (e-government, e-participation, and network readiness), and agriculture relevance are low. The business environment is also not at the same level as many of the other countries, though given that inflation is high (17.1%) and lending rates stand at 19.3 percent, it is only the third highest after Zimbabwe and Malawi. It is also ranked 177 of 190 countries in terms of ease of doing business (Department of Business and trade -UK 2023). This is mainly linked to an unstable macroeconomic stability that is external shocks

including oil price fluctuations.

Strong areas in Botswana, as depicted in Figure 5, include the business environment and subsequently positive score for adopter characteristics (education, income per capita, share of agricultural land, gender equality in education), prior digital penetration, and are all supported by the literature discussed previously. Botswana is among the most economically successful countries in Africa. Botswana has a stable political environment and is ranked 87th of 190 countries in ease of doing business. The country also has one of the highest per capita income in Africa (US \$ 6,621 in 2023), a competitive tax system, streamlined processes for licensing, business registration, and other regulatory requirements (Chandrawat and partners 2023), and averaging lending rate 10.780 percent per annum (CEIC Data 2023). Factors that all contribute positively to the business enabling environment and thus digital readiness.

Overall, Malawi ranks third from the bottom in terms of preparedness and readiness for digital upscaling (Figure 5). The most important factors that contribute to readiness in Malawi is the business environment. Essential infrastructure Relevance to agriculture and policy environment respond positively to the score. Prior digital penetration, and e-extension however are still low. These areas are critical and need investments to spur rapid and effective adoption of digital technology for climate adaptation. Malawi is reported to have wide macroeconomic imbalances because of unsustainable debt and thus eroding fiscal space (IMF 2023b). Its lending rate was 23 percent per annum, more than twice that of Botswana (CEIC Data 2023b). High lending rates generally are negative for the ease of doing business and affect the business enabling environment negatively.

Similarly, Mozambique is among the four least ready countries for digital adaptation. When the digital readiness pillars are compared for Mozambique alone, the business environment and the relevance of agriculture are contributing significantly to the overall readiness (Figure 5). It is important to note that the inflation rates in Mozambique stood at just below 3.5 percent in 2020, the openness to trade is the highest in the region, and the ease of doing business is also good. But prior digital penetration remains low, and human capital skills are among the region's lowest.

Namibia ranked second on the digital adaptation readiness index. Like the other countries, the business enabling environment scores highly as a main contributor to digital readiness as the business environment is better than in most countries of focus; essential infrastructure is relatively

available; and there is considerable prior digital penetration. The cost of essential complementary technology (mobile bundles) is, however, high in Namibia, the e-extension ranking is low, and the relevance of agriculture for the overall economy and jobs remains lowest in the Zambezi. Tanzania ranks as the third-most ready country for digital adaptation in the Zambezi region. It has the essential infrastructure, the adopter characteristics are favorable, and the business environment is also good when compared to Tanzania's own performance in the other seven pillars that define the overall index. However, low prior digital penetration and poor scores on e-extension contribute less to the overall index (Figure 5). Tanzania would therefore need to improve on e-extension services (e-government, e-extension, and network readiness) as well as access to computing equipment and internet usage in general.

A low-middle-income country, Zambia ranked 4th in digital adaptation readiness. While the low share of agriculture in GDP, poor prior digital penetration, and low e-extension scores pull the index downward, Zambia business enabling environment is the biggest contributor to digital readiness. Areas requiring urgent investments include e-government, e-extension, and digital infrastructure to improve network and access. Finally, Zimbabwe's strengths are its educated population (adopter characteristics), availability of essential infrastructure, and high relevance of agriculture to the economy. However, Zimbabwe's business enabling environment scores lower among all factors affecting its overall score on digital readiness.

Figure 6 confirms the importance and relevance of the business enabling environment in explaining digital readiness. In countries where this scores highly, the digital readiness index is also high. Only agricultural relevance appears to suggest that those countries that depend less on agriculture also have higher levels of readiness. This also makes sense because the country's dependence on agriculture wanes with economic transformation and, hence, with the availability of many of the factors that define the other six pillars.

In the Zambezi River Basin, countries rank differently in terms of access, use, and affordability of mobile telephony, ICT, and internet. They are also at different stages of embracing E-government. In general, Botswana, Namibia, and Tanzania rank high in the region on e-government, e-participation, and network readiness indicators, while Angola, Mozambique, and Malawi have lower scores and rank lowest in the region (Table 3). The affordability of internet as measured by the cost of data, and the cost of internet bundles

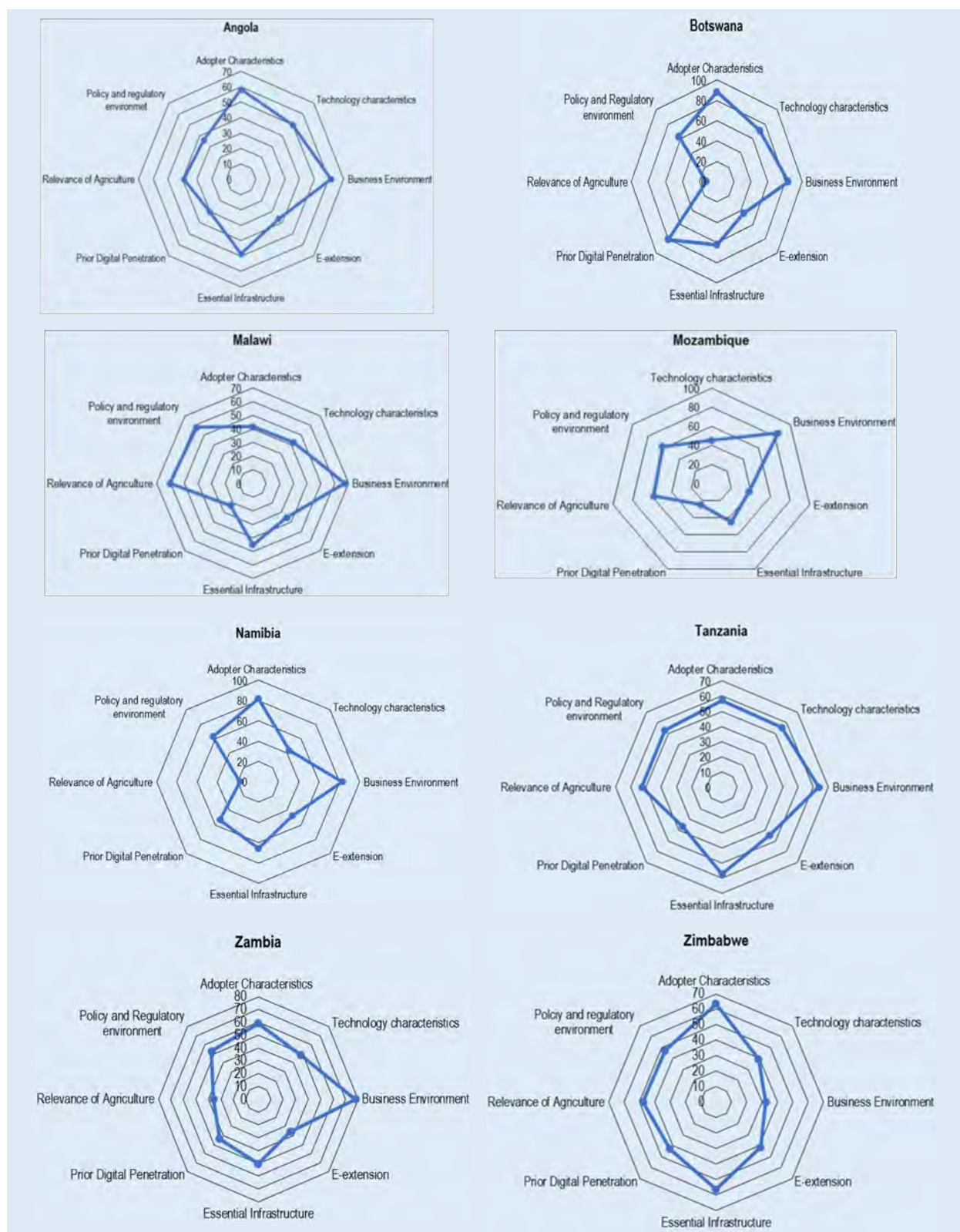


Figure 5 Factors contributing to the country-level digital adaptation readiness (Source: Authors)

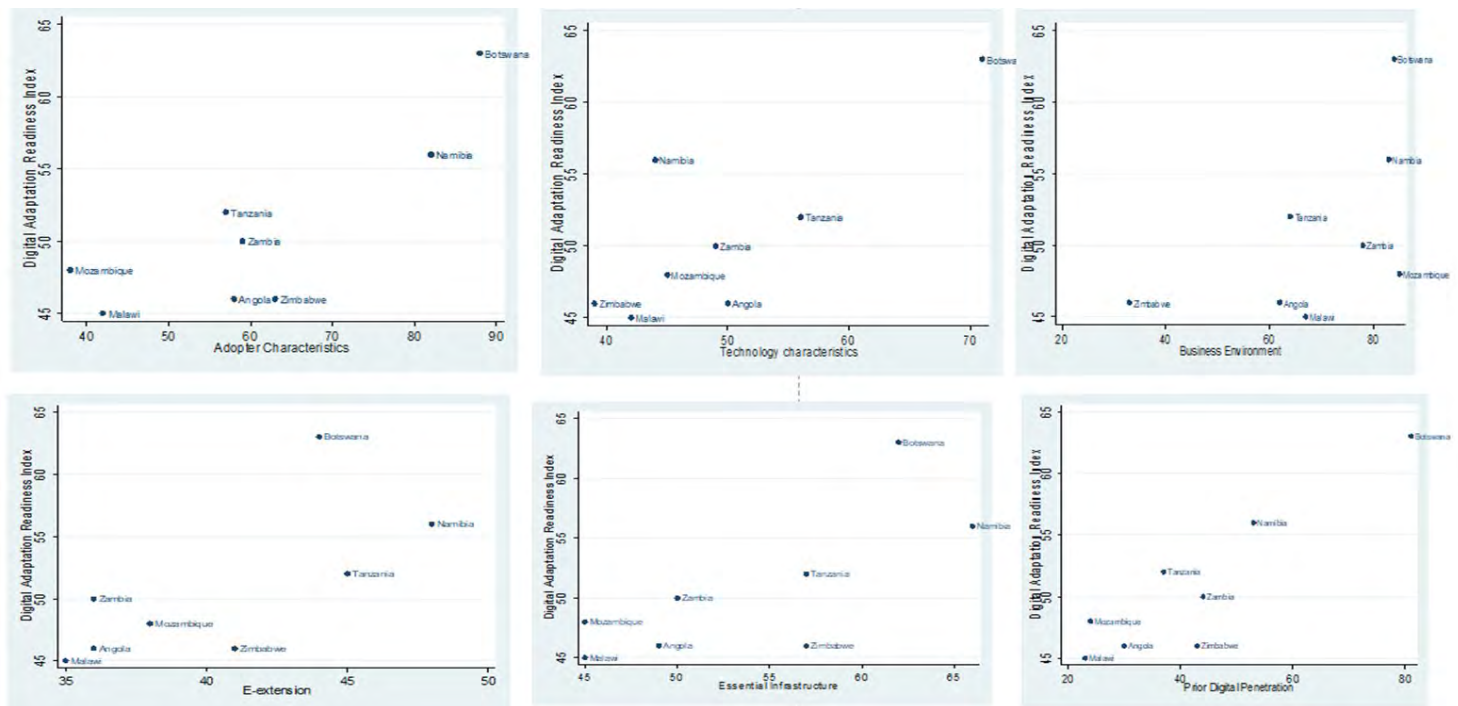


Figure 6 Correlations between pillars and the Digital Adaptation Index (Source: Authors)

as a share of income per capita also confirms the heterogeneity across countries. While Angola ranks low on e-government measures, it is among the top three (including Botswana and Namibia) countries where internet is affordable. However, mobile internet is also more expensive in Malawi, Mozambique, and Zimbabwe at the time of writing this paper.

The available data in ICT skills and in Internet penetration in homes shows that huge populations of the Zambezi countries have limited access to internet as well as computers. Malawi, Tanzania Mozambique, Zambia, and Angola have the least proportions of households with computers (below 10% of the population), whereas Botswana, Namibia, and Zimbabwe have comparatively higher proportions of the population using internet. Mobile infrastructure as proxied by mobile cellular coverage is high in Namibia and Botswana where the populations covered by mobile cellular network is close to 100 percent, but mobile broadband penetration is very low across all the Zambezi countries. For example, less than 10 percent of people in Angola, and Tanzania have active mobile broadband subscriptions.

So, there are significant challenges in access, use, and availability of infrastructure throughout the region, there is a gender gap in favor of men in respect of access to internet and

mobile phone ownership in some of the countries, and the overall E-Governance and E-Participation rates are among the lowest countries in the world. While these results show the need for intervention, it is important to also note that these challenges are heterogeneous and similarly the approaches for intervention need to emphasize different aspects of these challenges. In the following sections, these indicators are discussed alongside others for each of the Zambezi River Basin countries.

Figure 7 shows that countries with better enabling environment have a higher digital penetration rates. This is because private sector investments are easily attracted into the sector. Barriers to entry into the business are lower hence entry costs are reduced. Most countries need to come up with policies and strategies that will make it conducive for businesses to enter the costs and provide a service at an affordable rate. Where the environment is difficult, the costs are passed on to the users who reduce usage and hence penetration rates. Digital penetration is lowest in Mozambique and Malawi while Namibia and Botswana have highest penetration due to a better enabling environment.

Figure 8 reinforces the positive correlation between good enabling business environment and digital penetration and

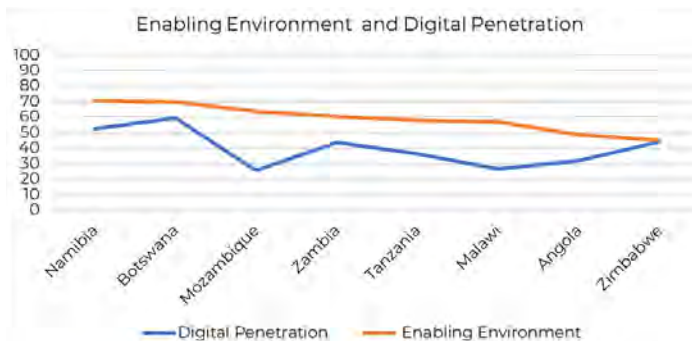


Figure 7 Enabling Environment and Digital Penetration (Source: Authors)

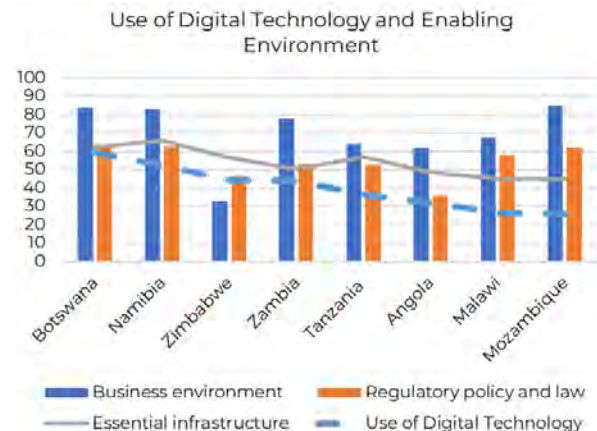


Figure 8 Use of Digital Technology and Enabling Environment (Source: Authors)

hence use of digital technologies. Zimbabwe business environment, also as emerged during the national consultations, is challenging due to several factors including high taxes and inflation. The country lags in terms of infrastructure to support use of digital technology. Essential infrastructure also plays a key role in the use of digital technology as it allows for better penetration. Botswana, Namibia, Mozambique, and Zambia have better business environments compared to other countries in the Zambezi River Basin. This has resulted in them acquiring better infrastructure leading to increased use of digital technologies. A challenging business environment has negative effects on business operations and hence business profitability. This results in few investments channeled to the sector consequently affecting the use of digital technologies.

5. Discussion

Based on the climate projection in the Zambezi River Basin, the impacts of climate change are expected to increase in intensity and frequency. There is, therefore, need for transformation towards climate-smart and resilient food systems through the adoption of digital solutions to climate change adaptation. The adaptation processes are a result of individuals' ability to implement adaptive decisions, thus transforming that capacity into action (functioning) (Nelson et al. 2007). In turn, the adaptation processes can lead to system transformations when new livelihood strategies are adopted (e.g., when climate-related disturbances force systems to depend on new, diversified livelihood options), as well as system adjustments, when systems are improved to reduce vulnerability and strengthen future adaptive capacity.

Addressing the various constraints and weaknesses while capitalizing on strengths at the country-specific and regional level would be extremely useful in driving technology adoption for climate adaptation and mitigation in the Zambezi River Basin. In general, these range from those seeking to improve infrastructure that supports digital technology (e.g., electricity, towers, weather stations, software for remote sensing, mobile phones, computers, etc.), the regulatory and policy environment (e.g., e-government, institutions), as well enhancing adopter attributes (e.g., skills development, use of cooperatives), and technology characteristics (e.g., costs, affordability, accessibility, user-friendliness for all genders, profitability). However, the ease of doing business will determine the level to which the countries will be ready to adopt digital technology even with such soft and middleware infrastructure boosts.

Likewise, findings presented in Section 4 show that an enabling environment is critical for uptake of digital solutions both in and outside the agricultural sectors. This is supported by the fact that in all countries under focus, a high enabling environment index correlates better with better digital technology readiness. This, however, does not negate the fact that other factors need to compliment the business enabling environment for adaptation to be seamless. For example, policy regulatory environment, prior infrastructure penetration, and adopter characteristic were also key for digital technology readiness. Zambia for example while ranked high in enabling policy and policy and regulatory environment is still falls in the mid-range for digital adaption readiness in comparison to Botswana, Namibia and Tanzania due to low or outdated infrastructure. Zimbabwe on the other

Table 3 Enabling environment and digital penetration in the Zambezi River Basin (Source: Authors' compilation from various cited sources)

Indicator	Angola	Botswana	Malawi	Mozambique	Namibia	Tanzania	Zambia	Zimbabwe
Population covered by a mobile-cellular network (%)	90	>95	86	85	>95	95	86.9	93.0
Mobile-cellular subscriptions per 100 inhabitants	45	162.6	47.8	49	102.105	85.75	103.918	88.8
Active mobile broadband subscriptions per 100 inhabitants	3.7	67.3	16.6	17	90	9.80	51.1	51.7
Individuals owning a mobile phone (%)	78.6			31.4		88	83.4	89.6
Male mobile phone ownership as a percentage of total male population (%)	77.6			56			86.5	
Female mobile phone ownership as a percentage of total female population (%)	79.8			26	54		80.8	
Individuals using the Internet, total (%)	36	64	15.50	17	40.5	20	19.0	25.1
Female internet users as a percentage of the total female population (%)				17	20		12.0	30.0
Male Internet users as a percentage of total male population (%)				27			17.3	28.9
Households with Internet access at home (%)		63.5	10.5	2		23	49.7	50.1
Households with a computer at home (%)	9.2	14.75	5.19	7.33	16.53	3.8	8.1	14.2
Individuals with basic ICT skills (%)			40				51.9	3.9
Affordability, out of 100	56 th (39.83)	13 th (71.1)	54 th (42.27)	51 st (45.17)	53 rd (43.51)	38 th (55.55)	48 th (48.54)	58 th (39.34)
Price per 1.5 GB (US\$)	4.65	5.68	6.70	2.86	8.45	6.17	2.82	11.22
Mobile cellular basket as a percentage of GNI per capita (%)	3.1	1.1	22.1	19.9	2	4.5	5.62	13.4
Mobile broadband basket as a percentage of GNI per capita (%)	3.2	1.1	20	8	2.5	6.8	3.7	23.5
E-Government Development Index (Rank out of 193 countries)	159 th (0.38)	115 th (0.53)	165 th (0.35)	163 rd (0.36)	104 th (0.57)	152 nd (0.42)	148 th (0.42)	126 th (0.50)
E-Participation Index (Rank out of 193 countries)	122 nd (0.45)	137 th (0.37)	129 th (0.42)	103 rd (0.52)	112 nd (0.5)	98 th (0.56)	158 th (0.31)	122 nd (0.45)
Network Readiness Index (Rank out of 134 countries)	126 th (25.99)	102 nd (38.03)	119 th (29.00)	125 th (26.55)	109 th (35.66)	107 th (35.83)	112 th (33.93)	122 nd (28.74)

hand scored a high index on adopter characteristics due to having adult literacy of over 95%. However, its overall index on readiness scored 8th due to low scores for business enabling environment and policy and regulatory environment. Even with a high score on essential infrastructure, readiness remains low. This underscores how critical a business enabling environment is for readiness for digital adoption under the agriculture sector and other sectors in general.

Still, it is important to note that all the countries in the

Zambezi River Basin have major opportunities to expand digital solutions as demand still outstrip supply of digital services in climate change adaptation. A lot of private organizations are actively involved in scaling the much-needed digital infrastructure. The countries under study have created enabling environments for ICT development through the removal of restrictions on entry in investing in the ICT sector as shown by the growth and rise in the number of telecommunication companies in those countries over time. The high number of active ICT companies offers a

competitive environment to improve the quality-of-service delivery (FAO and ITU 2022).

Macroeconomic stabilization efforts can also create an environment that attracts investors and development of ideal policies that can drive climate adaptation and mitigation. Currently, there is low budget allocation to climate change adaptation and digitalization in all the countries in this study (CGIAR 2019). Therefore, the private sector must step in. The governments can promote this through various ways including subsidies and tax rebates for those willing to co-finance climate interventions. Accessing funds such as the Green Climate Fund, the Governments can acquire and upgrade the weak digital infrastructure to a level where it is able to generate the information required by stakeholders in decision making. Climate financing allows vulnerable communities that would otherwise not afford various technologies to be able to access the required technologies. Climate financing can also be done through public-private or private-private partnerships, and these are key not only in generating funding but also in technical and innovation transfer that are lacking in smallholder communities.

6. Limitations and Caveats

While it would have been preferable to utilize panel data encompassing a wide range of households and businesses in the Zambezi river basin, such data is not available. Therefore, this study relied on valuable national-level data, data extracted from literature, and primary information gathered from experts, farmers, and other stakeholder groups.

7. Conclusion

The study aimed to document the factors within the enabling environment that are critical for digital technology readiness and upscaling in the Zambezi region. The International Water Management Institute (IWMI), in partnership with the Global Centre on Adaptation (GCA) and the African Development Bank (AFDB), conducted consultation meetings and literature reviews in the Zambezi riparian countries to identify the potential of digital technologies in accelerating climate adaptation in agriculture. The evidence gathered was used to prepare this paper.

The major finding was that a good business enabling environment promotes the development of digital infrastructure, which facilitates the adoption and use of digital technologies. Policies and regulations should support the entry and viable operations of private players in the digital sector. Countries with better business enabling environments, such as

Botswana, Namibia, Mozambique, and Zambia, have better infrastructure, policies, and higher penetration rates of digital technologies. Adoption and use of digital technologies are also high in these countries. On the other hand, countries with less enabling environments, such as Zimbabwe and Malawi, have low penetration rates and use of digital technologies.

While this study does not cover all the issues that hinder climate adaptation in the Zambezi region, it provides key insights into the nature of climate risks in the Zambezi River Basin countries, the ongoing adaptation efforts, and the constraints that challenge the upscaling of digital tools for climate adaptation. The enabling policy environment was identified as a key pillar for enhancing the adoption of digital technology, as it creates an environment for private, public, and development partners to invest. The findings indicate that a conducive policy environment reduces barriers and increases readiness for adaptation.

The proposed set of interventions is not exhaustive but is considered relevant and important for the respective countries, given the challenges they face and the observed threats, strengths, opportunities, and weaknesses. The research approach, which includes stakeholder consultations and secondary data analysis, provides advantages that would not be possible with either method alone. While the research results are useful for the purposes outlined, future research should consider adding more in-depth studies in sampled rural areas to track various heterogeneities within each country's locations and cultures. This would provide further insights but would also require more time and resources due to the diverse nature of the Zambezi River Basin and its eight riparian countries.

8. Recommendations

Successful implementation of digital climate adaptation requires the creation of a supportive business environment for digital technology. This can be achieved through relevant, well-harmonized, and inclusive policies. Such policies will encourage investment from the private sector, development partners, and other stakeholders. Governments should also aim to reduce barriers to entry into the sector, allowing more players to participate and lowering the costs of digital services. Additionally, efforts should be made to establish infrastructure in various areas within the Zambezi River Basin.

Countries like Botswana should prioritize stabilizing the macroeconomic environment, streamlining application processes, and reducing borrowing costs to foster a favorable

business environment. Angola, Malawi, Mozambique, Tanzania, and Zambia should address challenges related to low levels of digitization, strengthen E-government systems, and improve the regulatory environment. Malawi and Mozambique should also invest in training programs that focus on digital tools to enhance human capital skills.

While Namibia is relatively better prepared, it should address the issue of high data costs. Similarly, Zimbabwe, Tanzania, and Malawi need to find ways to reduce the high costs of data.

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References

- African Development Bank AfDB. 2021. Africa Adaptation Acceleration Program. Abidjan: AfDB. <https://www.afdb.org/en/topics-and-sectors/initiatives-partnerships/africa-adaptation-acceleration-program> (accessed on May 18, 2022)
- Beck, L.; Bernauer, T. 2011. How will combined changes in water demand and climate affect water availability in the Zambezi river basin? *Global Environmental Change* 21(3): 1061-1072. ISSN 0959-3780, <https://www.sciencedirect.com/science/article/pii/S0959378011000616>
- Beyer, M.; Wallner, M.; Bahlmann, L.; Thiemi, V.; Dietrich, J.; Billib, M. 2016. Rainfall characteristics and their implications for rain-fed agriculture: a case study in the Upper Zambezi River Basin. *Hydrological Sciences Journal* 61: 321-343
- Brazier, A. 2015. Climate Change in Zimbabwe, Facts for planners and decision makers. Berlin: Konrad-Adenauer-Stiftung. ISBN 0797467440, 9780797467446
- CEIC Data. 2023. Botswana Bank Lending Rate. <https://www.ceicdata.com/en/indicator/botswana/bank-lending-rate> (accessed on November 10, 2023)
- CEIC Data. 2023b. Malawi Bank Lending Rate. <https://www.ceicdata.com/en/indicator/malawi/bank-lending-rate#:~:text=Malawi%20Bank%20Lending%20Rate%20was,22.700%20%25%20pa%20for%20Aug%202023> (accessed on November 10, 2023)
- CGIAR. 2019. The Digitalization of African Agriculture Report 2018–2019. <https://cgispace.cgiar.org/bitstream/handle/10568/103198/Executive%20Summary%20V4.5%20ONLINE.pdf> (accessed on May 18, 2022)
- Chandrawat and Partners. 2023. Why Botswana? <https://www.chandrawatpartners.com/where-we-can-help/doing-business-in-botswana/#:~:text=Botswana%20is%20a%20desirable%20location,a%20generally%20favorable%20business%20environment> (accessed on November 20, 2023)
- Cornell University.; INSEAD.; WIPO. 2016: The Global Innovation Index 2016: Winning with Global Innovation. Ithaca, Fontainebleau, and Geneva.
- Department of Business and trade UK. 2023. Overseas business risk: Angola. <https://www.gov.uk/government/publications/overseas-business-risk-angola/overseas-business-risk-angola--2#:~:text=The%20last%20published%20World%20Bank,177%20out%20of%20190%20countries> (accessed on October 12, 2023).
- Devenish, A.J.M.; Schmitter, P.; Jellason, N.P.; Esmail, N.; Abdi, N.M.; Adanu, S.K.; Adolph, B.; Al-Zu'bi, M.; Amali, A.A.; Barron, J. 2023. One Hundred Priority Questions for the Development of Sustainable Food Systems in Sub-Saharan Africa. *Land* 12: 1879. <https://doi.org/10.3390/land12101879>
- Dinesh, D.; Rodriguez, A.; M. L. 2019. Scaling Digital Agriculture to Leapfrog Adaptation Pathways. <https://ccafs.cgiar.org/news/scaling-digital-agriculture-leapfrog-adaptation-pathways#:~:text=Digital%20agriculture%20involves%20using%20a,better%20informed%20decisions%20in%20Ghana> (accessed on June 01, 2022)
- FAO. 2015; World Food and Agriculture – FAO Statistical Pocketbook, Food and Agriculture org/explore/africa/zambia#:~:text=Some%20of%20Zambia's%20adaptation%20measures,drought%20and%20flood%20prone%20areas (accessed on August 11, 2022)
- FAO and ITU. 2022. Status of digital agriculture in 47 sub-Saharan African countries. Rome: FAO
- Fuglie, K.; Gautam, M.; Goyal, A.; Maloney, W. 2019. Improving the Enabling Environment for Technology Adoption. https://doi.org/10.1596/978-1-4648-1393-1_ch4
- Goh, L. 2022. How AgriTech is Transforming Traditional Agriculture in Emerging Markets: “Think Big, Act Fast, Start Small”. In: Kharas, H.; McArthur, J.; Ohno, I.; eds. Breakthrough: The Promise of Frontier Technologies for Sustainable Development. Brookings Institution Press, p. 254
- Hamududu, H. B.; Killingtveit, Å. 2016. Hydropower Production in Future Climate Scenarios; the Case for the Zambezi River. *Energies* 9(7): 502. <https://doi.org/10.3390/en9070502>
- Hamududu, H. B.; Ngoma, H. 2019. Impacts of Climate Change on Water Availability in Zambia: Implications for Irrigation Development. Feed the Future Innovation Lab for Food Security

- Policy. Research Paper 146.
- IMF. 2023. Botswana Financial System Stability Assessment. IMF Country Report No. 23/336. September 2023. <https://www.imf.org/en/Publications/CR/Issues/2023/09/21/Botswana-Financial-System-Stability-Assessment-539353> (accessed on November 30, 2023)
- IMF. 2023b. Malawi Financial System Stability Assessment. IMF Country Report No. 23/336. September 2023. <https://www.imf.org/en/Publications/CR/Issues/2023/09/21/Botswana-Financial-System-Stability-Assessment-539353> (accessed on November 30, 2023)
- IPCC. 2014. Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, R.K. Pachauri.; L.A. Meyer eds.]. Geneva, Switzerland: IPCC 151 pp
- Kararach, G.; Nhamo, G.; Mubila, M.; Nhamo, S.; Nhemachena, C.; Babu, S. 2017. Reflections on the Green Growth Index for developing countries: A focus of selected African countries. *Development Policy Review* 36: O432–O454. doi:10.1111/dpr.12265
- Kling, H.; Stanzel, P.; Preishuber, M.; 2014. Impact modelling of water resources development and climate scenarios on Zambezi River discharge. *Journal of Hydrology: Regional Studies* 1:17–43. <https://doi.org/10.1016/j.ejrh.2014.05.002>
- Malambo, M.; Tembo, M.C.; Chapoto, A.; Matchaya, G.; Kasoma-Pele, W.; Aheeyar, M.; Ebrahim, G.; Ajayi, O.C.; Afun-Ogidan, O.; Fakudze, B. 2023. Digital Adaptation in Agriculture Profile for Zambia. Colombo, Rotterdam and Abidjan: International Water Management Institute, Global Center on Adaptation and the African Development Bank.
- Mayoyo, A.; Chapoto, A.; Matchaya, G.; Aheeyar, M.; Chiwunze, G.; Ebrahim, G.; Ajayi, O.C.; Afun-Ogidan, O.; Fakudze, B.; Kasoma-Pele, W. 2023. Digital Climate Adaptation in Agriculture Profile for Zimbabwe. Colombo, Rotterdam, and Abidjan: International Water Management Institute, Global Centre on Adaptation, and the African Development Bank.
- McKinsey and Company. 2020. Agriculture's connected future: How technology can yield new growth. <https://www.mckinsey.com/industries/agriculture/our-insights/agricultures-connected-future-how-technology-can-yield-new-growth> (Accessed on September 20, 2022)
- Midgley, S.J.E.; Davies, R.A.G.; Chesterman, S. 2011. Climate Risk and Vulnerability Mapping in Southern Africa: Status quo 2008 and future 2050. For the Regional Climate Change Programme for Southern Africa RCCP, UK Department for International Development DFID. Cape Town: OneWorld Sustainable Investments.
- Mtewa, R.P. 2018. Assessing Socio-Economic Impacts of Drought and Coping Mechanisms: A Case Study on Musikavanhu Area, Chipinge District, Zimbabwe. Free State: University of the Free State
- Nardo, M.; Saisana, M.; Saitelli, A.; Tarantola, S. 2005. Tools for Composite Indicators Building. European Commission Joint Research Centre. Italy: Institute for the Protection and Security of the Citizen Econometrics and Statistical Support to Antifraud Unit
- Nhemachena, C.; Matchaya, G.; Nhemachena, C.R.; Karuaihe, S.; Muchara, B.; Nhlengethwa S. 2018. Measuring Baseline Agriculture-Related Sustainable Development Goals Index for Southern Africa. *Sustainability* 10(3): 849. <https://doi.org/10.3390/su10030849>
- Nelson, D. R.; Adger, W. N.; Brown, K. 2007. Adaptation to Environmental Change: Contributions of a Resilience Adaptation to Environmental Change: Contributions of a Resilience Framework. *Annual Review of Environment and Resources* 32 (November).
- Ngoma, H.; Lupiya, P.; Kabisa, M. 2021. Impacts of climate change on agriculture and household welfare in Zambia: an economy-wide analysis. *Climatic Change* 169 (27). <https://link.springer.com/article/10.1007/s10584-021-03168-z>
- POTRAZ. 2021. Abridged Postal and Telecommunications Sector Performance Report <https://www.techzim.co.zw/wp-content/uploads/2021/12/Abridged-Sector-Performance-Report-Q3-2021-HMed.pdf> (accessed on November 23, 2023)
- Sachs, J.; Schmidt-Traub, G.; Durand-Delacre, D. 2016. Preliminary Sustainable Development Goal SDG Index and Dashboard. SDSN Working Paper.
- Saisana, M.; Tarantola, S. 2002. State-of-the-Art Report on Current Methodologies and Practices for Composite Indicator Development. Report EUR 20408 EN. ISPRA: European Commission-Joint Research Centre. <https://www.eea.europa.eu/data-and-maps/indicators/economic-water-productivity-of-irrigated/state-of-the-art-report>
- Sanchez.; G. S. 2018. The Zambezi River Basin: water resources management. Energy-Food-Water nexus approach, Stockholm: Stockholm University, Department of Geography.
- Schmidt-Traub, G.; Kroll, C.; Teksoz, K. 2017. National baselines for the Sustainable Development Goals assessed in the SDG Index and Dashboards. *Nature Geosci* 10: 547–555. <https://doi.org/10.1038/ngeo2985>
- Senzanje, T.L.; Dirwai. 2020. Water and agriculture in the Zambezi River Basin. https://aquaknow.jrc.ec.europa.eu/en/system/files/Annex30_Senzanje_UnKWZN_FinalReport.pdf
- Statista. 2020. Ease of doing business in Zimbabwe in 2020, by topic. [https://www.statista.com/statistics/1245753/ease-of-doing-business-in-zimbabwe-by-topic/#:~:text=Zimbabwe%](https://www.statista.com/statistics/1245753/ease-of-doing-business-in-zimbabwe-by-topic/#:~:text=Zimbabwe%20)

- 20scored%2055%20points%20in, scale%20of%20zero%20to%20100 (Accessed on November 26, 2023).
- Stephenson, J.; Chellew, T.; von Kockritz, L.; Rose, A.; Dinesh, D. 2021. Digital agriculture to Enable Adaptation: A supplement to the UNFCCC NAP Technical Guidelines. CGIAR Research Program on Climate Change, Agriculture and Food Security CCAFS, Volume Working Paper No. 372
- Trendov, N.; M.; Varas, S.; Zeng, M. 2019. Digital Technologies In Agriculture and Rural Areas Status Report. Rome: Food and Agriculture Organization of the United Nations.
- UNDP. 2022. Climate change adaptation. New York: United Nations Development Programme.
- UNDP. 2016. The United Nations Development Programme: Human Development Report 2016. Technical Report. New York: UNDP.
- USAID. 2015. Climate Change Information Fact Sheet Southern Africa. https://www.climatelinks.org/sites/default/files/asset/document/Southern%20Africa%20Climate%20Info%20Fact%20Sheet_FINAL.pdf (Accessed on September 13, 2022).
- World Bank. 2010. The Zambezi River Basin: A Multi-Sector Investment Opportunities Analysis - Basin Development Scenarios. Washington DC: World Bank. <https://openknowledge.worldbank.org/handle/10986/2959>
- World Bank. 2018. Zimbabwe Climate Risk Country Profile. Washington DC: World Bank. www.worldbank.org
- World Bank. 2019. Climate Change Knowledge Portal Zimbabwe: Impacts on Agriculture. Washington DC: World Bank.
- World Bank. 2019b. Zimbabwe Country Profile; World Bank Climate Change Portal; https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/14956-WB_Zimbabwe%20Country%20Profile-WEB%20%281%29.pdf (Accessed on September 13, 2022).
- World Bank. 2020. Zimbabwe - Climatology | Climate Change Knowledge Portal. <https://climateknowledgeportal.worldbank.org/country/zimbabwe/climate-data-historical> (Accessed on September 13, 2022).
- World Bank. 2022. World Bank Development Indicators. Washington DC: World Bank; <https://data.worldbank.org/> (Accessed on November 14, 2023)
- Zambezi River Authority. 2007. Integrated Water Resource Management Strategy for the Zambezi River Basin. Gaborone and Harare: SADC and Zambezi River Authority
- ZimStat ; POTRAZ. 2020. ICT Household Survey 2020, ZimStat and POTRAZ; http://www.zimstat.co.zw/wpcontent/uploads/publications/Economic/Ict/ICT_Survey_Household_2020.pdf (Accessed on November 14, 2023).